

MATH 301: Homework 3

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February 25, 2026

Section 2.3: q4

Let $S_4 := \left\{ 1 - \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}$. Find $\inf S_4$ and $\sup S_4$.

Let $f(n) := 1 - \frac{(-1)^n}{n}$.

Consider the first few terms in the sequence:

$$\begin{aligned} f(1) &= 1 - \frac{(-1)^1}{1} = 1 + 1 = 2 \\ f(2) &= 1 - \frac{(-1)^2}{2} = 1 - \frac{1}{2} = \frac{1}{2} \\ f(3) &= 1 - \frac{(-1)^3}{3} = 1 + \frac{1}{3} = \frac{4}{3} \\ f(4) &= 1 - \frac{(-1)^4}{4} = 1 - \frac{1}{4} = \frac{3}{4} \\ &\dots \end{aligned}$$

We can reconstruct this sequence into two subsequences based on whether the index is odd or even. Let K be the set of odd natural numbers and J be the set of even natural numbers:

$$K := \{k \in \mathbb{N} : k \text{ is odd}\} \quad J := \{j \in \mathbb{N} : j \text{ is even}\}$$

For odd $k \in K$, $(-1)^k = -1$, giving the function $g(k)$:

$$g(k) = 1 + \frac{1}{k}$$

For even $j \in J$, $(-1)^j = 1$, giving the function $h(j)$:

$$h(j) = 1 - \frac{1}{j}$$

Consider the first terms of $g(k)$ where $k \in \{1, 3, 5, \dots\}$:

$$g(1) = 2, \quad g(3) = \frac{4}{3}, \quad g(5) = \frac{6}{5}, \quad \dots$$

Observe that $g(k)$ is strictly decreasing. For $x \in K$:

$$\begin{aligned} x &< x + 2 \\ \implies \frac{1}{x} &> \frac{1}{x + 2} \\ \implies 1 + \frac{1}{x} &> 1 + \frac{1}{x + 2} \\ \implies g(x) &> g(x + 2) \end{aligned}$$

Consider the first terms of $h(j)$ where $j \in \{2, 4, 6, \dots\}$:

$$h(2) = \frac{1}{2}, \quad h(4) = \frac{3}{4}, \quad h(6) = \frac{5}{6}, \quad \dots$$

Observe that $h(j)$ is strictly increasing. For $z \in J$:

$$\begin{aligned}
 & z < z + 2 \\
 \implies & \frac{1}{z} > \frac{1}{z + 2} \\
 \implies & -\frac{1}{z} < -\frac{1}{z + 2} \\
 \implies & 1 - \frac{1}{z} < 1 - \frac{1}{z + 2} \\
 \implies & h(z) < h(z + 2)
 \end{aligned}$$

Because $g(k)$ is strictly decreasing, its supremum is its first term (2). Because $h(j)$ is strictly increasing, its infimum is its first term ($\frac{1}{2}$). Therefore, for the entire set S_4 :

$$\therefore \sup S_4 = 2, \quad \inf S_4 = \frac{1}{2}$$

Section 2.3: q5

Find the infimum and supremum, if they exist, of each of the following sets.

- (a) $A := \{x \in \mathbb{R} : 2x + 5 > 0\}$, (b) $B := \{x \in \mathbb{R} : x + 2 \geq x^2\}$,
(c) $C := \{x \in \mathbb{R} : x < 1/x\}$, (d) $D := \{x \in \mathbb{R} : x^2 - 2x - 5 < 0\}$.

Part (a): $A := \{x \in \mathbb{R} : 2x + 5 > 0\} \implies \inf A = \frac{-5}{2}, \nexists \sup A$

Part (b): $B := \{x \in \mathbb{R} : x + 2 \geq x^2\} \implies \inf B = -1, \sup B = 2$

Part (c): $C := \{x \in \mathbb{R} : x < 1/x\} \implies \nexists \inf B, \sup C = 1$

Part (d): $D := \{x \in \mathbb{R} : x^2 - 2x - 5 < 0\} \implies \inf D = 1 - \sqrt{6}, \sup D = 1 + \sqrt{6}$

Section 2.3: q6

Let $S \subseteq \mathbb{R}, S \neq \emptyset$ and bounded below. Prove that $\inf S = -\sup\{-s : s \in S\}$

S is bounded below $\implies \exists u$ s.t. $u := \inf S$

$u := \inf S \implies \forall s \in S, (u \leq s) \wedge (u \text{ is the largest lowerbound})$

u is the largest lowerbound $\implies$ If w is a lowerbound. Then $w \leq u$

Let $|S| = C, C \subseteq \mathbb{N}$ Consider ordering of $S := \{s_1 \leq s_2 \leq \dots \leq s_n \leq \dots\}$

Define $T := \{-s : s \in S\}$ Observe $u := \inf S \implies u \leq s_1$

Consider ordering of $T := \{-s_1 \geq -s_2 \geq \dots \geq -s_n \geq \dots\}$

S is bounded below $\implies -S$ is bounded above $\implies \exists \sup -S$

Observe $u :=$ a lower bound $S \implies -u \geq -s_1$

Thus $-u \geq -s_1 \geq -s_2 \geq \dots \geq -s_n \geq \dots \implies \forall x \in -S, -u \geq x \implies -u$ is an upperbound of S

Assume $\exists (z < u)$ s.t. $\forall x \in -S, z \geq x$ $z := (-u + \frac{1}{n}), n \in \mathbb{N}$

$$\implies -u + \frac{1}{n} > -s_1 \implies u - \frac{1}{n} < s_1$$

$$\implies \sup S = u$$

Section 2.3: q8

Let $S \subseteq \mathbb{R}, S \neq \emptyset$. Show $u \in \mathbb{R}$ is an upper bound of $S \iff (t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S)$

Prove: $u \in \mathbb{R}$ is an upper bound of $S \implies ((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S))$

Suppose for the sake of contradiction: $(t \in S) \wedge (t \in \mathbb{R}) \wedge (t > u)$

$u \in \mathbb{R}$ is an upper bound of $S \implies \forall s \in S, u \geq s$

$(t \in S) \wedge (t > u) \implies u$ is not an upperbound of $S \nmid$

$\therefore (u \in \mathbb{R} \text{ is an upper bound of } S) \implies ((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S))$

Prove: $((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S)) \implies (u \in \mathbb{R} \text{ is an upper bound of } S)$

Consider the contrapositive:

$(u \in \mathbb{R} \text{ is NOT an upper bound of } S) \implies \exists t \text{ s.t. } ((t \in \mathbb{R}) \wedge (t > u) \wedge (t \in S))$

If $u \in \mathbb{R}$ is NOT an upper bound of $S \implies \exists t \in (S \cup \mathbb{R}) \ni t > u \implies ((t \in \mathbb{R}) \wedge (t > u) \wedge (t \in S))$

Section 2.3: q10

Show that if A and B are bounded subsets of $\mathbb{R}$. Then $A \cup B$ is a bounded set.

Show that $\sup(A \cup B) = \sup\{\sup A, \sup B\}$

Let $u := \sup S, v := \sup B, w := \sup\{u, v\}$

$(x \in A \implies x \leq u \leq w) \wedge (x \in B \implies x \leq v \leq w)$

$\implies w$ is an upper bound of $A \cup B$

$z := \text{any upper bound of } A \cup B \implies z$ is an upper bound of A and B

$\implies u \leq z \wedge v \leq z \implies w \leq z$

$\therefore w = \sup(A \cup B)$

Section 2.4: q2

If $S := \{\frac{1}{n} - \frac{1}{m} | n, m \in \mathbb{N}\}$ find $\inf S$ and $\sup S$

Lemma 1: Consider the subproblem $\lim_{n \rightarrow \infty} \frac{1}{n}$ where n is an arbitrary value $n \in \mathbb{N}$

By backwards search:

$$\forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N}$$

$$\forall n \geq N(\epsilon), \left| \frac{1}{n} - 0 \right| < \epsilon$$

$$\equiv \frac{1}{n} < \epsilon$$

$$\equiv n > \frac{1}{\epsilon} \text{ if } n \geq N(\epsilon)$$

$$\leftarrow N(\epsilon) > \frac{1}{\epsilon}$$

$\leftarrow$ Archimedean Property

Forward Logic:

$$\text{Observe } \frac{1}{\epsilon} \in \mathbb{R}$$

$\implies$ Archimedean Property Holds

$$\implies \exists N(\epsilon) \in \mathbb{N} \text{ s.t. } N(\epsilon) > \frac{1}{\epsilon}$$

$$\implies \forall n \in N(\epsilon), n > \frac{1}{\epsilon}$$

$$\implies \frac{1}{n} < \epsilon \implies \left| \frac{1}{n} - 0 \right| < \epsilon$$

$$\therefore \frac{1}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Thus by (Lemma 1) for an arbitrary natural number, we know $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Lemma 2: Consider $\frac{1}{n}$ is strictly decreasing.

$$P(n) := \frac{1}{n} > \frac{1}{n+1}$$

$$P(1) := \frac{1}{1} > \frac{1}{2} \text{ holds}$$

$$\text{Suppose } P(k) := \frac{1}{k} > \frac{1}{k+1}$$

$$\implies k < k+1$$

$$\implies k+1 < k+2 \implies \frac{1}{k+1} > \frac{1}{k+2}$$

Therefore, we can say $\frac{1}{n}$ is strictly decreasing.

Thus by (Lemma 2) $\forall A := \{\frac{1}{a} : a \in \mathbb{N}\}, \sup(A) = 1$

Since $n, m \in \mathbb{N}$, we know $\frac{1}{n} \leq 1$ and $-\frac{1}{m} < 0$. Therefore, $\frac{1}{n} - \frac{1}{m} < 1$, making 1 an upper bound.

Apply the same logic to qualify -1 as a lower bound.

In summary:

$$\begin{aligned} n \in \mathbb{N} &\implies 0 < \frac{1}{n} \leq 1 \\ m \in \mathbb{N} &\implies -1 \leq -\frac{1}{m} < 0 \end{aligned}$$

Finding Supremum and Infimum

Find the maximum value of $\frac{1}{n}$ and minimum $\frac{1}{m}$.

$$\text{Considering the bounds (Lemma 1 \& Lemma 2): } \left\{ \begin{array}{l} \implies \frac{1}{n} = 1 \\ \implies -\frac{1}{m} \approx 0 \end{array} \right.$$

Consider: $\lim_{m \rightarrow \infty} (1 - \frac{1}{m})$

$$|(1 - \frac{1}{m}) - 1| < \epsilon \iff |-\frac{1}{m} + (1 - 1)| < \epsilon \iff |-\frac{1}{m}| < \epsilon \iff |\frac{1}{m}| < \epsilon \iff \frac{|1|}{m} < \epsilon$$

$$\leftarrow m > \frac{1}{\epsilon} \leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{m \rightarrow \infty} (1 - \frac{1}{m}) = 1$$

Consider: $\lim_{n \rightarrow \infty} (\frac{1}{n} - 1)$

$$|(\frac{1}{n} - 1) - (-1)| < \epsilon \iff |\frac{1}{n} + (1 - 1)| < \epsilon \iff |\frac{1}{n}| < \epsilon \iff \frac{|1|}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$

$$\leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{n \rightarrow \infty} (\frac{1}{n} - 1) = -1$$

Therefore we can say the $\sup S = 1$ and $\inf S = -1$

Section 2.4: q7

Let A, B be bounded nonempty subsets of $\mathbb{R}$, and let $A + B := \{a + b | a \in A, b \in B\}$.
Prove $\sup(A + B) = \sup A + \sup B$ and $\inf(A + B) = \inf A + \inf B$.

Consider for the following:

$$\sup A = a_s \implies \forall a \in A, a < a_s$$

$$\sup B = b_s \implies \forall b \in B, b < b_s$$

Observe:

$$\begin{array}{l} (a < a_s) \\ + (b < b_s) \\ \hline a + b < a_s + b_s \end{array}$$

Thus we can say $a_s + b_s$ is an upperbound. To prove it is the least upper bound, let $\epsilon > 0$. By the properties of suprema, no number strictly smaller than the supremum can be an upper bound. Therefore, there must exist some $a \in A$ and some $b \in B$ such that:

$$a > a_s - \frac{\epsilon}{2}$$

$$b > b_s - \frac{\epsilon}{2}$$

Adding these two inequalities together yields:

$$a + b > a_s + b_s - \epsilon$$

Since we found an element $a + b \in A + B$ that is strictly greater than $a_s + b_s - \epsilon$ for any arbitrary $\epsilon > 0$, no number smaller than $a_s + b_s$ can be an upper bound.

Therefore, $\sup(A + B) = a_s + b_s = \sup A + \sup B$.

Observe, $a + b$ represent arbitrary elements of $A + B$

Therefore, $\sup A + B = a_s + b_s$

$\inf A = a_i \implies \forall a \in A, a > a_i$

$\inf B = b_i \implies \forall b \in B, b > b_i$

Observe:

$$\begin{array}{r} (a > a_i) \\ + (b > b_i) \\ \hline a + b > a_i + b_i \end{array}$$

Thus we can say $a_i + b_i$ is a lowerbound.

Observe, $a + b$ represent arbitrary elements of $A + B$

Therefore, $\inf A + B = a_i + b_i$

To prove it is the greatest lower bound, let $\epsilon > 0$. By the properties of infima, no number strictly larger than the infimum can be a lower bound. Therefore, there must exist some $a \in A$ and some $b \in B$ such that:

$$a < a_i + \frac{\epsilon}{2}$$

$$b < b_i + \frac{\epsilon}{2}$$

Adding these two inequalities together yields:

$$a + b < a_i + b_i + \epsilon$$

Since we found an element $a + b \in A + B$ that is strictly less than $a_i + b_i + \epsilon$ for any arbitrary $\epsilon > 0$, no number larger than $a_i + b_i$ can be a lower bound.

Therefore, $\inf(A + B) = a_i + b_i = \inf A + \inf B$.

Section 2.4: q9

Let $X = Y := \{x \in \mathbb{R} : 0 < x < 1\}$. Define $h : X \times Y \rightarrow \mathbb{R}$ by $h(x, y) := 2x + y$.

(a) For each $x \in X$, find $f(x) := \sup\{h(x, y) : y \in Y\}$; then find $\inf\{f(x) : x \in X\}$.

(b) For each $y \in Y$, find $g(y) := \inf\{h(x, y) : x \in X\}$; then find $\sup\{g(y) : y \in Y\}$.

Compare with the result found in part (a).

Item (a)

$$\forall x \in X, f(x) := \sup\{h(x, y) : y \in (0, 1)\} \implies f(x) = 2x + 1$$

$$f(x) = 2x + 1 \wedge x \in (0, 1) \implies \inf\{h(x, y) : x \in X\} = 2(0) + 1 = 1$$

Item (b)

$$\forall y \in Y, g(y) := \inf\{h(x, y) : x \in (0, 1)\} \implies g(y) = 2(0) + y = y$$

$$g(y) = y \wedge y \in (0, 1) \implies \sup\{g(y) : y \in Y\} = 1$$

Section 2.4: q14

If $y > 0$ show $\exists n \in \mathbb{N} \ni \frac{1}{2^n} < y$

Recall:

$$a > b \implies \frac{1}{a} < \frac{1}{b}$$

Observe:

$$P(n) : n < 2^n$$

$$P(1) : 1 < 2^1$$

$$P(k) : k < 2^k$$

$$2k < 2^{k+1}$$

$$2k + 1 < 2k < 2^{k+1}$$

$$\therefore P(n) \text{ holds } \forall n \in \mathbb{N}$$

Thus we can say $\frac{1}{2^n} < \frac{1}{n}$. By Archimedian Property as $y > 0 \implies \frac{1}{n} \leq y$

$$\frac{1}{2^n} < \frac{1}{n} \leq y$$

Section 2.4: q15

Modify the argument in Theorem 2.4.7 to show that $\exists y > 0 \in \mathbb{R}$ s.t. $y^2 = 3$

Theorem 2.4.7 $\exists x > 0 \in \mathbb{R} \ni x^2 = 2$

$$S = \{s \in \mathbb{R} \mid s \geq 0 \text{ and } s^2 < 3\}$$

Existence of Supremum

$$0^2 = 0 < 3 \implies 0 \in S \implies S \neq \emptyset$$

Let $\forall s \in S, s < 3$ Suppose $\exists s \in S \ni s \geq 3 \implies s^2 \geq 9 > 3 \nmid \implies S$ is bounded above.

$S \neq \emptyset \wedge S$ is bounded above $\implies \exists x \in \mathbb{R} \ni x = \sup S$ (By Completeness Property.)

Trichotomy

By Trichotomy, one and only one, of the following must be true:

$$(x^2 < 3) \quad (x^2 > 3) \quad (x^2 = 3)$$

(Case: $(x^2 > 3)$)

Suppose for the sake of contradiction x is the least upper bound of S

$$\text{Show: } \forall n \in \mathbb{N}, (x - \frac{1}{n})^2 > 3 \implies \forall s \in S, (x - \frac{1}{n})^2 > s^2 \implies (x - \frac{1}{n}) > s$$

Observe the following:

$$(x - \frac{1}{n})^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2}$$
$$x^2 - \frac{2x}{n} + \frac{1}{n^2} \xrightarrow{\text{Dropping } \frac{1}{n^2}} (x - \frac{1}{n})^2 > x^2 - \frac{2x}{n}$$

To hold supposition, the inequality must be strictly greater than three.

$$x^2 - \frac{2x}{n} > 3$$
$$x^2 - 3 > \frac{2x}{n}$$
$$\frac{1}{x^2 - 3} < \frac{n}{2x}$$
$$\frac{2x}{x^2 - 3} < n$$

Since $x^2 > 3 \implies (x^2 - 3) > 0 \implies \exists n \ni x > \frac{1}{n} > 0$.

$$(x - \frac{1}{2})^2 > 3 \implies \forall s \in S, s^2 < 3 < (x - \frac{1}{n})^2$$

$$\implies x - \frac{1}{n} \text{ is an upper bound of } S$$

$$x - \frac{1}{n} < x \implies x \text{ is not the least upper bound of } S \nmid$$

$$\therefore (x^2 \neq 3)$$

(Case: $(x^2 < 3)$)

Observe the following:

$$(x + \frac{1}{n})^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2}$$

Note $\forall n \in \mathbb{N}, \frac{1}{n^2} \leq \frac{1}{n}$

$$x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}$$

$$\implies x^2 + \frac{2x+1}{n} < 3$$

$$\implies \frac{2x+1}{n} < 3 - x^2$$

$$\implies \frac{n}{2x+1} > \frac{1}{3-x^2}$$

$$\implies n > \frac{2x+1}{3-x^2}$$

As $x^2 < 3 \implies 3 - x^2 > 0$. Therefore by Archimedian Property $\implies (x + \frac{1}{n}) \in S$
 $\implies x$ is not an upperbound $\nless$.

$\therefore (x^2 \nless 3)$

As $x^2 \nless 3$ and $x^2 \nless 3$. By Trichotomy $x^2 = 3$